QS-VC MCU-Based Video Conferencing — User & Operations Manual

Quantum-Safe Enterprise Video Conferencing — MCU (Multipoint Control Unit) Edition Version 1.0.0 | March 2026

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1. Introduction & Overview

QS-VC MCU Edition is the **enterprise/intranet variant** of the QS-VC Quantum-Safe Video Conferencing Platform. It is designed for **secured deployments** within organizations such as banks, government agencies, defence establishments, and large enterprises that require:

- **Private network operation** (MPLS VPN / IPSec VPN / LAN) — no public internet
- **Legacy endpoint support** (SIP/H.323 room systems — Polycom, Cisco, etc.)
- **Server-side video composition** — single composite stream per participant
- **Bandwidth efficiency** — MCU transcodes and composes centrally
- **On-premise recording** with tamper-proof audit logging
- **Quantum-safe encryption** (CRYSTALS-Kyber-1024 + Dilithium-5)
- **Air-gapped deployment** capability

Key Capabilities

Feature	Description
Video Composition	Grid, speaker, presentation, spotlight, and filmstrip layouts
Audio Mixing	Multipoint audio bridge with server-side mixing

Transcoding	H.264/H.265/VP8/VP9/AV1 ↔ any codec conversion
SIP Gateway	Connect SIP-based room systems and desk phones
H.323 Gateway	Connect legacy H.323 video conferencing hardware
RTMP Ingest	Live stream to external platforms
Composite Recording	Server-side MP4 recording of composed video
500 Participants/Room	Up to 500 participants per MCU room
100 Concurrent Rooms	Up to 100 simultaneous MCU rooms per instance
Quantum-Safe E2EE	Kyber-1024 + X25519 hybrid key exchange

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2. MCU vs SFU — When to Use MCU

Aspect	SFU (SaaS Mode)	MCU (Enterprise Mode)
Architecture	Selective Forwarding Unit — forwards streams as-is	Multipoint Control Unit — transcodes & composes
Client bandwidth	High (receives N streams)	Low (receives 1 composite stream)
Client CPU	High (decodes multiple streams)	Low (decodes single stream)
Server CPU	Low (forward only)	High (transcode + compose)
Legacy devices	WebRTC only	SIP, H.323, any codec
Recording	External composition needed	Native composite recording
Network	Public internet + TURN	Private MPLS VPN / LAN
Meeting initiation	Shareable links (URL)	Corporate calendar / directory
Guest access	Yes	No — corporate identity only
Best for	Small groups, modern browsers, SaaS	Large rooms, room systems, secure environments

Decision Matrix

Condition	Recommended Mode
Participants on different networks (home/office/mobile)	**SFU**
All participants on same corporate MPLS VPN	**MCU**
Need to connect Polycom/Cisco room systems	**MCU**
Bandwidth-constrained MPLS links	**MCU**

Compliance requires on-premise recording	**MCU**
Meeting size > 50 participants	**MCU** (or SFU cascade)
Internet-facing SaaS product	**SFU**

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3. System Architecture

MCU Mode Architecture

```
┌──────────────┐      MPLS VPN / L3VPN      ┌──────────────┐
│ Endpoint A   │ ─── SRTP ──────────────► │              │
│ (Branch HQ)  │                          │   MCU        │
└──────────────┘                          │ (On-Premise) │
                                          │              │
┌──────────────┐      MPLS VPN / L3VPN     │ • Transcoding│
│ Endpoint B   │ ─── SRTP ──────────────► │ • Composition│
│ (Data Center)│                          │ • Recording  │
└──────────────┘                          │ • AI (local) │
                                          └──────────────┘
┌──────────────┐      MPLS VPN / L3VPN           │
│ Endpoint C   │ ─── SRTP ──────────────────────┘
│ (Branch Delhi)│
└──────────────┘


No TURN/STUN needed — all endpoints on same private network
No NAT traversal — direct IP connectivity within VPN
```

Service Architecture (Intranet Mode)

```
┌──────────────────────────────────────────────────┐
│                   FRONTEND                       │
│        React + Vite (Port 5173)                  │
│        MCU_ENABLED=true, MODE=intranet           │
└──────────────┬───────────────────────────────────┘
               │
    ┌──────────┼──────────┐
    │          │          │
┌───▼───┐  ┌───▼───┐  ┌───▼───┐
│  Sig  │  │  MCU  │  │  Web  │
│ :4001 │  │ :4002 │  │ :5173 │
│       │  │       │  │       │
│WebSock│  │GStream│  │ Vite  │
│JSON-  │  │Compos.│  │ Dev   │
│RPC 2.0│  │SIP/H. │  │Server │
│       │  │323 GW │  │       │
└───┬───┘  └───┬───┘  └───────┘
    │          │
┌───▼──────────▼──────────────────────────┐
│       PostgreSQL (5432) + Redis (6379)  │
```

```
│                + MinIO (9000)                    │
└──────────────────────────────────────────────────┘
```

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4. Prerequisites & Hardware Requirements

Software Requirements

Component	Minimum	Recommended
Node.js	v20+	v24+
Docker	v20+	v29+
OS	Windows 10/11, Ubuntu 20+, RHEL 8+	Windows 11, Ubuntu 22+
Browser	Chrome 90+, Edge 90+	Latest Chrome/Edge
GStreamer	— (dev mode uses abstraction)	v1.22+ (production)

Hardware Requirements

Small Deployment (50 concurrent meetings, 500 users)

Component	Specification
Application Server	32 vCPU, 128 GB RAM, 2 TB NVMe SSD, 2× 10GbE NIC
AI Appliance	16 vCPU, 64 GB RAM, 1 TB NVMe, 2× NVIDIA A10 GPU
Storage Server	8 vCPU, 32 GB RAM, 20 TB RAID-10

Medium Deployment (200 concurrent meetings, 2000 users)

Component	Specification
Application Servers	3× clustered K8s nodes
AI Appliances	2× load-balanced Triton servers
Storage Servers	2× replicated MinIO + PostgreSQL
Load Balancer	1× F5 / HAProxy
HSM	1× Thales Luna / AWS CloudHSM on-prem

Large Deployment (1000+ concurrent meetings, 10000+ users)

Component	Specification
Application Servers	10× full K8s cluster
Dedicated SFU/MCU Servers	5× bare-metal, 10GbE
AI Appliances	4× A100 GPUs, clustered Triton
Storage Servers	3× Ceph/MinIO distributed

PostgreSQL HA	2× Patroni cluster
Redis Sentinel	2× clusters
Kafka	3-node cluster
HSMs	2× active-passive
Network	25GbE spine-leaf fabric

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5. Installation & Setup

Development Setup

Step 1: Clone the Repository

```
git clone <repository-url> QS_VC
cd QS_VC
```

Step 2: Install Dependencies

```
npm install
```

This installs dependencies for all workspaces including the MCU service (`@qsvc/mcu`).

Step 3: Start Docker Infrastructure

Ensure Docker Desktop is running, then:

```
docker compose up -d postgres redis minio
```

Service	Port	Purpose
PostgreSQL 16	5432	Database
Redis 7	6379	Sessions & pub/sub
MinIO	9000 / 9001	Recording storage (S3-compatible)

> ***MinIO Console***: *http://localhost:9001* *Login:* `qsvc_minio` / `qsvc_minio_secret`

Step 4: Verify Installation

```
# Run all tests
npx vitest run --reporter=verbose
```

Step 5: Start MCU Platform

```
node start-intranet.js
```

This single command starts all three MCU-mode services sequentially. See Section 7 for details.

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6. Configuration Reference

MCU Service Configuration

The MCU service is configured via environment variables and defaults defined in `services/mcu/src/index.ts`:

Parameter	Default	Description
`MCU_PORT`	`4002`	MCU HTTP API port
`maxRooms`	`100`	Maximum concurrent MCU rooms
`maxParticipantsPerRoom`	`500`	Maximum participants per room
`defaultLayout`	`speaker`	Default video composition layout
`outputResolution`	`1920×1080`	Composite output resolution
`outputFps`	`30`	Composite output frame rate
`outputVideoBitrate`	`4000 kbps`	Composite video bitrate
`outputAudioBitrate`	`128 kbps`	Composite audio bitrate
`outputVideoCodec`	`h264`	Output video codec
`outputAudioCodec`	`opus`	Output audio codec
`recordingEnabled`	`true`	Enable server-side recording
`recordingPath`	`./recordings`	Recording output directory
`sipEnabled`	`true`	Enable SIP gateway
`sipPort`	`5060`	SIP signaling port
`h323Enabled`	`true`	Enable H.323 gateway
`h323Port`	`1720`	H.323 control port
`rtmpEnabled`	`true`	Enable RTMP ingest/streaming

SIP/H.323 Gateway Configuration

Configured in `services/mcu/src/sip-gateway-config.ts`:

Parameter	Default	Description
`sipListenPort`	`5060`	SIP UDP/TCP port
`sipTlsPort`	`5061`	SIP TLS port
`sipDomain`	`sip.qsvc.local`	SIP domain / registrar
`sipTransport`	`tls`	Default transport (udp/tcp/tls/ws/wss)

`sipTrunkEnabled`	`true`	Enable PSTN SIP trunking
`sipTrunkProvider`	`twilio`	SIP trunk provider
`h323ListenPort`	`1720`	H.323 TCP port
`h323GatekeeperMode`	`discover`	Gatekeeper mode (none/discover/required)
`rtpPortRange`	`20000-30000`	RTP media port range
`srtpEnabled`	`true`	Enable SRTP encryption
`dtmfMode`	`rfc2833`	DTMF signaling method

Supported Video Codecs

`H.264` , `H.265` , `VP8` , `VP9` , `AV1` , `H.263`

Supported Audio Codecs

`Opus` , `G.722` , `G.711 μ-law (PCMU)` , `G.711 A-law (PCMA)` , `AAC-LD`

Intranet-Mode Environment Variables

When launched via `start-intranet.js` , these variables are automatically set:

Variable	Value	Target Service
`CORS_ORIGINS`	`*`	Signaling
`MCU_ENABLED`	`true`	Signaling
`MCU_URL`	`http://localhost:4002`	Signaling
`VITE_MCU_ENABLED`	`true`	Web Frontend
`VITE_MCU_URL`	`http://localhost:4002`	Web Frontend
`VITE_DEPLOYMENT_MODE`	`intranet`	Web Frontend

---

7. Starting the MCU Platform

One-Command Launch (Recommended)

```
node start-intranet.js
```

This launches the enterprise MCU platform with a single command. The script:

1. **Starts Signaling Server** (port 4001) — WebSocket signaling with MCU mode enabled
2. **Starts MCU Service** (port 4002) — GStreamer video composition + SIP/H.323 gateway
3. **Starts Vite Web UI** (port 5173) — React frontend in intranet mode

Expected Output

```
╔═══════════════════════════════════════════════════════╗
║   QS-VC — Enterprise Intranet Deployment (MCU Mode)  ║
║   🏢 On-Premise • SIP/H.323 • Quantum-Safe E2EE     ║
║   MCU Video Composition + Audio Mixing               ║
╚═══════════════════════════════════════════════════════╝


━━━
```

```
cd services\mcu && npx tsx watch src/index.ts

# Terminal 3 — Web Frontend
cd frontend\web && npx vite --host 0.0.0.0
```

Verifying Service Health

```
# MCU Health (shows capabilities)
curl http://localhost:4002/health

# Expected response:
# {
#   "status": "ok",
#   "service": "qsvc-mcu",
#   "rooms": 0,
#   "capabilities": {
#     "videoCodecs": ["h264","h265","vp8","vp9","av1"],
#     "audioCodecs": ["opus","pcma","pcmu","g722","aac"],
#     "maxParticipants": 500,
#     "layouts": ["grid","speaker","presentation","spotlight","filmstrip"],
#     "sipEnabled": true,
#     "h323Enabled": true,
#     "rtmpEnabled": true,
#     "recordingEnabled": true
#   }
# }

# Signaling Health
curl http://localhost:4001/health

# SIP/H.323 Gateway Status
curl http://localhost:4002/api/gateway/status
```

---

8. User Guide — Joining Meetings

For WebRTC Participants (Browser / Desktop App)

MCU (Secured) meeting flow differs from SaaS:

Step	SaaS (Link Sharing)	MCU (Enterprise)
1	User clicks "New Meeting"	Admin creates meeting in corporate calendar (Exchange/Lotus)
2	Shares link via email/chat	Calendar invite sent to internal participants only
3	Anyone with link joins	Participants click calendar link on corporate device
4	Browser → SFU (internet)	Native app → MCU via VPN (no browser required)
5	Host ends meeting	Meeting ends per schedule or host action

6	Recording to cloud S3	Recording to on-premise MinIO, transcript to PostgreSQL

Secured Meeting Lifecycle

```
1. SCHEDULE → Admin creates meeting in corporate calendar (Exchange/Lotus)
2. INVITE   → Calendar invite sent to participants (internal only)
3. JOIN     → Participants click calendar link on corporate device
              → App connects to MCU via VPN (no browser — native app)
              → SSO + device certificate validated
4. MEDIA    → H.323/SIP/WebRTC via MCU (MPLS VPN, no internet)
5. END      → Meeting ends per schedule or host action
6. ARTIFACT → Recording saved to on-premise MinIO
              → Transcript saved to on-premise PostgreSQL
              → Audit log written (tamper-proof, hash-chained)
```

Authentication in Secured Mode

Layer	Implementation
Network	Only corporate MPLS VPN IPs can reach the MCU
Device	Mutual TLS with device certificates (issued by internal CA)
Identity	SAML 2.0 / OIDC via corporate IdP (AD FS / Keycloak)
Session	JWT with Dilithium-5 signature, 1-hour expiry
Meeting	Per-meeting RBAC (host, participant) from directory group

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9. User Guide — SIP/H.323 Endpoints

Connecting Room Systems

The MCU service includes a built-in SIP/H.323 gateway that bridges legacy video conferencing hardware to QS-VC meetings.

SIP Dial-In

1. On your SIP endpoint (e.g., Polycom Group 500), navigate to **Place a Call**
2. Enter the SIP URI:

   ```
   sip:<meeting-code>@sip.qsvc.local
   ```

3. The MCU will answer and bridge you into the meeting room
4. You will see the composite video layout (all participants in one view)

H.323 Dial-In

1. On your H.323 endpoint (e.g., Cisco TelePresence SX80), dial:

   ```
   sip.qsvc.local##<meeting-code>
   ```

   Or dial `sip.qsvc.local` and enter the conference ID when prompted.
2. The MCU will bridge you into the meeting

PSTN Dial-In

For audio-only phone participation:

1. Call any of the configured dial-in numbers:

Country	Number
India	+91-11-4000-XXXX
India (Toll-free)	+91-1800-XXX-XXXX
US	+1-646-XXX-XXXX
UK	+44-20-XXXX-XXXX
Singapore	+65-XXXX-XXXX
UAE	+971-4-XXX-XXXX

2. Enter the conference ID followed by `#`
3. DTMF controls:
   - `*6` — Mute/unmute
   - `*9` — Raise/lower hand

Connection Types & Auto-Codec Selection

Connection Type	Auto-Selected Video Codec	Auto-Selected Audio Codec
WebRTC	VP9	Opus
SIP	VP9	PCMA (G.711 A-law)
H.323	H.264	PCMA (G.711 A-law)
RTMP	VP9	Opus

The MCU automatically transcodes between codecs, so endpoints with different codecs can communicate seamlessly.

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10. User Guide — Meeting Controls & Layouts

Video Composition Layouts

The MCU composes all participant videos into a single output stream. Available layouts:

Layout	Description	Best For
Grid	Equal-size tiles in a grid pattern	Team meetings, all-hands
Speaker	Active speaker large, others in small strip	Presentations, discussions
Presentation	Screen share large, participants in side strip	Training, demos
Spotlight	Single speaker full screen	Executive addresses

Filmstrip	Horizontal strip at bottom of screen	Webinars, panels
Custom	Admin-defined custom layout	Special events

Grid Position Calculation

In grid layout, participants are arranged automatically:

- The grid columns = √(number of participants), rounded up
- Each participant is assigned a `(row, col)` position
- When participants join or leave, the grid recalculates automatically

Active Speaker Detection

- The MCU monitors audio levels from all participants
- The participant with the highest audio level is automatically flagged as the **active speaker**
- In `speaker` layout, the active speaker appears in the large central area
- Active speaker changes are announced to all participants

Changing Layout (Admin/Host)

Hosts can change the composition layout via the REST API:

```
curl -X POST http://localhost:4002/api/mcu/rooms/<roomId>/layout \
  -H "Content-Type: application/json" \
  -d '{"layout": "grid"}'
```

---

11. Recording & Live Streaming

Server-Side Composite Recording

MCU recording captures the composed video output (all participants in one view), unlike SFU recording which captures individual streams.

Start Recording

```
curl -X POST http://localhost:4002/api/mcu/rooms/<roomId>/recording/start
```

Response:

```
{
  "recording": true,
  "path": "./recordings/MEETING-CODE_1711155600000.mp4"
}
```

Stop Recording

```
curl -X POST http://localhost:4002/api/mcu/rooms/<roomId>/recording/stop
```

Recording files are saved in the configured `recordingPath` directory (`./recordings` by default) as MP4 files.

Recording Format

Property	Value
Container	MP4
Video Codec	H.264
Video Resolution	1920×1080 (configurable)
Video FPS	30
Video Bitrate	4000 kbps
Audio Codec	Opus (or AAC for compatibility)
Audio Bitrate	128 kbps

Live Streaming (RTMP)

Stream the meeting to an external platform (YouTube Live, Facebook Live, custom RTMP server):

```
curl -X POST http://localhost:4002/api/mcu/rooms/<roomId>/stream \
  -H "Content-Type: application/json" \
  -d '{"rtmpUrl": "rtmp://live.youtube.com/stream-key-here"}'
```

The GStreamer pipeline composites all participants and pushes the output via RTMP.

---

12. Admin Guide — MCU Management

Viewing All Active Rooms

```
curl http://localhost:4002/api/mcu/rooms
```

Response:

```
{
  "rooms": [
    {
      "id": "uuid-1",
      "meetingCode": "BOARD-MTG-001",
      "participantCount": 12,
      "layout": "speaker",
      "recording": true
    }
  ]
}
```

Viewing Room Details

```
curl http://localhost:4002/api/mcu/rooms/<roomId>
```

Response includes:

- Room ID, meeting code, layout
- All participants (name, connection type, audio/video status, grid position)
- Recording status
- Composition state (resolution, FPS, active speaker)
- SIP dial-in number and conference ID
- Room uptime

Removing a Participant

```
curl -X DELETE http://localhost:4002/api/mcu/rooms/<roomId>/participants/<peerId>
```

Setting Active Speaker Manually

```
curl -X POST http://localhost:4002/api/mcu/rooms/<roomId>/active-speaker \
  -H "Content-Type: application/json" \
  -d '{"peerId": "participant-id"}'
```

Gateway Status

Monitor the SIP/H.323/RTMP gateway:

```
curl http://localhost:4002/api/gateway/status
```

Response:

```
{
  "sip": {
    "enabled": true,
    "port": 5060,
    "protocol": "SIP/TLS",
    "registrar": "sip.qsvc.local",
    "supportedCodecs": ["opus", "pcma", "pcmu", "g722"],
    "activeCalls": 0
  },
  "h323": {
    "enabled": true,
    "port": 1720,
    "gatekeeper": "gk.qsvc.local",
    "supportedCodecs": ["h264", "h263", "pcma", "pcmu"],
    "activeCalls": 0,
    "supportedEndpoints": [
      "Polycom HDX/VVX/Group/Trio",
      "Cisco TelePresence/SX/MX/DX",
      "Tandberg Edge/Codec",
      "Lifesize Icon/Cloud",
```

```
      "Avaya Scopia",
      "Huawei TE/CE Series",
      "StarLeaf",
      "PeopleLink Ultra/Sky/Blaze"
    ]
  },
  "rtmp": {
    "enabled": true,
    "ingestUrl": "rtmp://mcu.qsvc.local/live",
    "activeStreams": 0
  }
}
```

---

13. API Reference

MCU Service (Port 4002)

Method	Endpoint	Description
GET	/health	Service health + capabilities
POST	/api/mcu/rooms/join	Join or create an MCU room
GET	/api/mcu/rooms	List all active rooms
GET	/api/mcu/rooms/:roomId	Get room details
POST	/api/mcu/rooms/:roomId/layout	Change composition layout
POST	/api/mcu/rooms/:roomId/active-speaker	Set active speaker
POST	/api/mcu/rooms/:roomId/recording/start	Start composite recording
POST	/api/mcu/rooms/:roomId/recording/stop	Stop recording
POST	/api/mcu/rooms/:roomId/stream	Start RTMP live stream
DELETE	/api/mcu/rooms/:roomId/participants/:peerId	Remove participant
GET	/api/gateway/status	SIP/H.323/RTMP gateway status

Join Room — Request Body

```
{
  "meetingCode": "BOARD-MTG-001",
  "peerId": "user-uuid-123",
  "displayName": "Dr. Sharma",
  "connectionType": "webrtc"  // or "sip", "h323", "rtmp"
}
```

Join Room — Response

```json
{
  "roomId": "room-uuid",
  "participantId": "user-uuid-123",
  "layout": "speaker",
  "compositionUrl": "/api/mcu/rooms/room-uuid/stream",
  "sipDialIn": "+91-1234567890",
  "sipConferenceId": "abc12345",
  "compositionState": {
    "outputResolution": { "width": 1920, "height": 1080 },
    "fps": 30,
    "activeSpeakerId": null,
    "presenterId": null
  }
}
```

Change Layout — Request Body

```json
{
  "layout": "grid"  // grid | speaker | presentation | spotlight | filmstrip | custom
}
```

Signaling Service (Port 4001)

Protocol	Endpoint	Description
WebSocket	`ws://localhost:4001/ws`	JSON-RPC 2.0 signaling
HTTP	`/health`	Health check

---

14. Security & Encryption

Zero Trust Security Architecture

The QS-VC MCU platform implements a 5-layer security model:

Layer	Implementation
Identity & Access	PQC Identity (Keycloak + Dilithium), MFA/FIDO2, RBAC + ABAC
Transport (Q-TLS)	Hybrid: X25519 + Kyber-1024 key exchange, Ed25519 + Dilithium-5 signatures
Media Encryption	SFrame E2EE with per-frame AES-256-GCM, Kyber-wrapped keys
Data at Rest	AES-256-GCM encrypted recordings, TDE for PostgreSQL, encrypted MinIO
Audit & Compliance	Tamper-proof logs (Merkle tree), SIEM integration, compliance dashboards

Quantum-Safe Cryptography

Algorithm	Use Case	Standard

CRYSTALS-Kyber-1024	Key encapsulation / key exchange	NIST FIPS 203
CRYSTALS-Dilithium-5	Digital signatures (JWT, certs, audit logs)	NIST FIPS 204
X25519	Classical key exchange (hybrid fallback)	RFC 7748
Ed25519	Classical signatures (hybrid fallback)	RFC 8032
AES-256-GCM	Symmetric encryption (media, data at rest)	NIST SP 800-38D
SHA-3-256	Hash function (audit log chain)	NIST FIPS 202

Key Rotation Schedule

Key Type	Rotation Interval	Method
TLS Certificates	90 days	ACME auto-renewal
TURN Secrets	1 hour	Vault dynamic secrets
JWT Signing Keys	24 hours	Dual-key overlap period
Meeting Keys (E2EE)	60 seconds	HKDF epoch derivation
Recording DEKs	Per-recording	Unique key per recording
Database TDE	30 days	Online re-encryption
Root CA	5 years	HSM ceremony

Compliance

Regulation	Status
ISO 27001	ISMS implementation, annual audit
IT Act 2000 (India)	LEA interface (court-ordered), audit logs
DPDP Act (India)	Data localization, consent, DPO dashboard
CCA (India)	PQC certificate compliance
GDPR (EU)	Data minimization, right to erasure
SOC 2 Type II	Continuous monitoring

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15. Network & Firewall Configuration

Required Ports (MCU Mode)

Port	Protocol	Service	Direction
`4001`	TCP	Signaling (WebSocket)	Inbound
`4002`	TCP	MCU REST API	Inbound

`5060`	UDP/TCP	SIP Signaling	Inbound
`5061`	TCP	SIP TLS	Inbound
`1720`	TCP	H.323 Control	Inbound
`5173`	TCP	Web UI (dev)	Inbound
`20000-30000`	UDP	RTP Media (SIP/H.323)	Inbound/Outbound
`5432`	TCP	PostgreSQL	Internal only
`6379`	TCP	Redis	Internal only
`9000/9001`	TCP	MinIO API / Console	Internal only

Network Topology for MCU Mode

```
Enterprise MPLS VPN / Private WAN
├── Branch HQ ──────────── MCU Server (Data Center)
├── Branch Office A ────── MCU Server
├── Branch Office B ────── MCU Server
├── Branch Office C ────── MCU Server
└── Remote VPN Users ───── MCU Server (via IPSec VPN)

No TURN/STUN needed — all endpoints on same private network
No NAT traversal — direct IP connectivity within VPN
No public internet traffic — all media stays within corporate WAN
```

---

16. Air-Gapped / On-Premise Deployment

For fully air-gapped environments with **no internet connectivity**:

Installation Workflow

```
1. MEDIA PREPARATION (at QS-VC facility)
   • Build all container images (Docker)
   • Package as OCI bundles on encrypted USB/HDD
   • Include: K8s binaries, Helm charts, AI model files
   • Sign all packages with Dilithium digital signature
   • Include offline Helm repo + container registry

2. HARDWARE SETUP (at customer site)
   • Rack & stack servers per hardware specification
   • Configure network (VLAN isolation, firewall rules)
   • Initialize HSM with key ceremony (split key holders)

3. SOFTWARE DEPLOYMENT
   • Boot installer from encrypted media
   • Verify digital signatures on all packages
   • Deploy private container registry (Harbor)
   • Push container images to private registry
```

```
   • Deploy K8s cluster (kubeadm / K3s / RKE2)
   • Apply Helm charts for all services
   • Initialize databases with schema + seed data
   • Load AI models into Triton Inference Server

4. CONFIGURATION
   • Tenant setup (organization name, branding, policies)
   • LDAP/AD integration (Keycloak LDAP federation)
   • TLS certificate installation (internal CA)
   • TURN server configuration (internal IPs — if needed)
   • Recording storage path configuration
   • SIP/H.323 gateway configuration

5. VALIDATION
   • Automated smoke test suite
   • Media quality test (internal video calls)
   • SIP/H.323 endpoint connectivity test
   • AI feature validation (captions, translation)
   • Security scan (CIS benchmarks)
   • Performance baseline measurement

Total Installation Time: 4–8 hours (automated)
```

---

17. Production Deployment (Docker)

Step 1: Configure Environment

```
cp .env.production.example .env.production
```

Edit `.env.production` with production values:

- Domain name
- Strong passwords for PostgreSQL, Redis, MinIO
- Secure JWT secret
- TURN secret
- SFU announced IP

Step 2: SSL Certificates

Place certificates in `./certs/` :

- `fullchain.pem` — Full certificate chain
- `privkey.pem` — Private key

Step 3: Deploy

```
docker compose -f docker-compose.prod.yml up -d
```

This starts the full production stack with:

- **Nginx** reverse proxy with TLS termination (ports 80/443)

- **PostgreSQL 16** with health checks
- **Redis 7** with AOF persistence and password auth
- **MinIO** object storage with health checks
- **coturn** TURN/STUN server
- **SFU** (mediasoup) with host networking
- **Signaling** server
- **Auth, Meeting, Recording** services
- **Web** frontend

---

18. Troubleshooting

Common Issues

Issue	Solution
MCU service won't start	Check port 4002 is free: `netstat -ano | findstr :4002`
SIP endpoint can't connect	Verify SIP port 5060 is open, check firewall rules
H.323 endpoint can't connect	Verify port 1720 open, check gatekeeper config
No audio from SIP endpoint	Check RTP port range (20000–30000) is open for UDP
Video quality is poor	Check `outputVideoBitrate` in MCU config, verify network bandwidth
Recording fails to start	Verify `recordingPath` directory exists and is writable
"Maximum MCU rooms reached"	Increase `maxRooms` config or close inactive rooms
Participant can't join	Check `maxParticipantsPerRoom` limit, verify authentication
Layout not changing	Verify correct `roomId` in API call, check room exists
PowerShell script blocked	Use `cmd /c "command"` or `Set-ExecutionPolicy RemoteSigned`
Docker not starting	Ensure Docker Desktop is running and fully initialized

Health Check Commands

```
# MCU Health
curl http://localhost:4002/health

# Gateway Status
curl http://localhost:4002/api/gateway/status

# Active Rooms
curl http://localhost:4002/api/mcu/rooms

# Signaling Health
curl http://localhost:4001/health
```

Logs

All services log to stdout. When using `start-intranet.js`, logs are prefixed with timestamps and service names:

```
[12:30:15] [Signaling] ✓ Signaling service is ready
[12:30:17] [MCU] ✓ MCU service is ready (GStreamer compositor + SIP/H.323 bridge)
[12:30:20] [Vite] ✓ Vite dev server is ready
```

Graceful Shutdown

Press `Ctrl+C` in the terminal running `start-intranet.js` — all services will be stopped cleanly.

---

19. Appendix — Supported Endpoints

Polycom

Model	Max Resolution	SIP	H.323	Dual Stream	Encryption
HDX 7000/8000/9000	4K	✅	✅	✅ (BFCP)	SRTP
VVX 500/600	1080p	✅	✅	✅	SRTP
Group 300/500/700	4K	✅	✅	✅	SRTP
Trio 8500/8800	1080p	✅	✅	✅	SRTP
Studio X30/X50/X70	4K	✅	✅	✅	SRTP
G7500	4K	✅	✅	✅	SRTP

Cisco

Model	Max Resolution	SIP	H.323	Dual Stream	Encryption
TelePresence SX10/20/80	4K	✅	✅	✅ (BFCP)	SRTP
MX200/300/700/800	4K	✅	✅	✅	SRTP
Room Kit/Mini/Plus	4K	✅	✅	✅	SRTP
Board 55/85	4K	✅	✅	✅	SRTP
Desk Pro	4K	✅	✅	✅	SRTP
DX70/DX80	4K	✅	✅	✅	SRTP

PeopleLink

Model	Max Resolution	SIP	H.323	Dual Stream	Encryption
Ultra HD	1080p	✅	✅	✅	SRTP
Sky 100/200	1080p	✅	✅	✅	SRTP
Blaze 300/400/500	1080p	✅	✅	✅	SRTP
Ivory 100S	1080p	✅	✅	✅	SRTP

Auro 500	1080p	✅	✅	✅	SRTP

Lifesize

Model	Max Resolution	SIP	H.323	Dual Stream	Encryption
Icon 300/450/500/700	4K	✅	✅	✅	SRTP
Cloud	4K	✅	✅	✅	SRTP

Avaya

Model	Max Resolution	SIP	H.323	Dual Stream	Encryption
Scopia XT5000	1080p	✅	✅	✅	H.235
B179	1080p	✅	✅	✅	H.235
IX Meeting Server	1080p	✅	✅	✅	H.235

Huawei

Model	Max Resolution	SIP	H.323	Dual Stream	Encryption
TE30/40/50/60	4K	✅	✅	✅	SRTP
CE200/CE400	4K	✅	✅	✅	SRTP

QS-VC — Quantum-Safe Enterprise Video Conferencing Platform (MCU Edition) Built for Banks, Government, Defence & Large Enterprises Quantum-Safe E2EE • SIP/H.323 Interop • On-Premise • Air-Gapped Ready